

Scalability & Future Plan

Phase 8 — Project Demonstration

This document is a ready-to-use guide for presenting PocketSmart AI: what to prepare, what to say, what to click, and how to handle problems.

1. Demo Objective

Show that PocketSmart AI turns a budget and a few requirements into a complete, shoppable plan, and that the application is secure, modular and ready to grow.

Duration: about 8 to 10 minutes including questions.

2. Before the Demo — Checklist

- Python environment and packages installed.
- .env contains a valid GOOGLE_API_KEY and your own SECRET_KEY.
- Server starts with `python -m uvicorn app:app --reload` and opens at `http://127.0.0.1:8000`.
- Internet connection tested; Gemini API responds.
- One sample outfit photo saved on the desktop.
- Browser zoom set so text is readable on the projector.
- A backup ready: screenshots or a screen recording, in case the internet or API fails.
- Remember: the app stores data in memory, so do not restart the server after creating your demo account.
- Register the demo account once before the audience arrives if you want to save time.

3. Suggested Speaking Roles

Roles are up to the team; a simple split is:

- **Nandhini.M** — introduction and closing.
- **Amrin.J** — registration, login and security.
- **Pooja Shri.R** — Home and Party planners.
- **Yakshitha.M** — Jewelry planner, history and future work.

4. Step-by-Step Demo Flow

Step 1 — Introduction (1 minute)

- Say: "Planning a budget for a home, a party or jewelry usually means many tabs and a lot of guessing. PocketSmart AI does it in one place."
- Show: landing page.

Step 2 — Register and log in (1 minute)

- Do: register a new user, then log in.

- Say: passwords are hashed with bcrypt; the login token lives in an HTTP-only cookie and expires after 30 minutes.
- Show: redirect to the dashboard.

Step 3 — Home planner (2 minutes)

- Do: enter budget 100000, 5 lights, 3 fans, 2 furniture pieces, 1 dining table; select Living Room and Kitchen; submit.
- Say: the app builds a prompt for Gemini, asks for JSON only, then adds shopping links.
- Show: categories, item prices, remaining budget, suggestions, and click one Amazon or IKEA link.

Step 4 — Party planner (2 minutes)

- Do: budget 50000, 40 guests, Birthday, Hall, with catering, decoration and entertainment.
- Say: this planner covers services as well as products.
- Show: cost split and links to Swiggy, Zomato and BookMyShow.

Step 5 — Jewelry planner (2 minutes)

- Do: budget 30000, occasion Wedding, preference "gold, traditional", upload the outfit photo.
- Say: the image is sent to Gemini, which describes the outfit's colours and style before suggesting jewelry.
- Show: outfit analysis, recommendations, styling tips and Tanishq or CaratLane links.

Step 6 — Dashboard and History (1 minute)

- Do: return to the dashboard, then open the history page.
- Show: the latest five recommendations and the full history.

Step 7 — Logout and security (30 seconds)

- Do: log out, then try to open /dashboard.
- Show: access is denied.

Step 8 — Behind the scenes (optional, 1 minute)

- Show: the folder structure in the editor and the /docs page listing all routes.

Step 9 — Limitations and future work (1 minute)

- Say: data is in memory now; next steps are a database, the latest Gemini SDK, tighter CORS, automated tests and deployment.

5. Key Points to Emphasise

- One app, three planners, one consistent experience.
- Structured AI output makes link building and history possible.
- Security basics are in place: hashing, tokens, blacklist, protected routes.
- Clean separation between routes, authentication, AI logic and models.
- Honest about limits: estimates are not live prices.

6. Possible Questions and Suggested Answers

How accurate are the prices?

They are AI estimates for planning. The shopping links let the user check real prices immediately.

How do you make sure the AI stays within budget?

The prompt instructs it to. In this version the total is not verified in code; adding that check is on our roadmap.

What happens if the AI returns something that is not JSON?

The code strips markdown fences and tries to extract the first JSON block. If that fails, an error is returned; better error handling is planned.

Is user data safe?

Passwords are hashed and never stored in plain text, and tokens expire and are blacklisted at logout. For production we would add a database, HTTPS, restricted CORS and rate limiting.

Why not use shop APIs?

Search links need no partnerships or keys and work immediately. APIs could be added later for live prices.

Why in-memory storage?

It kept the prototype simple. The storage code is small and can be replaced by a database.

Can it work for other cities or currencies?

The prompts are written for India and INR, but the structure would work for other regions by changing the prompt and platform lists.

7. Troubleshooting During the Demo

- **Page does not load** — check the terminal; restart the server (then register again because memory is cleared).
- **AI request fails or times out** — check internet and API key; use the backup screenshots.
- **Model or SDK error** — the code uses the older google.generativeai package with gemini-1.5-flash; update to the current SDK and a supported model before the demo.
- **Login stops working** — the token may have expired (30 minutes) or the server restarted; log in again.
- **Image upload fails** — use a normal JPG or PNG.

8. Materials to Include in This Folder

- screenshots/ — one screenshot for each step: landing page, register, login, dashboard, each planner result, history.
- A screen recording of the full flow (optional but recommended).
- Demo video link: *add link here*.
- Presentation slides: *add file or link here*.

9. Screenshot Checklist

- Landing page
- Registration page
- Login page
- Dashboard (with recent items)

- Home planner form and result
- Party planner form and result
- Jewelry planner form, uploaded image and result
- History page
- Access denied after logout

10. Closing Statement (Sample)

"PocketSmart AI shows how AI can move from giving advice to helping people act. It plans the budget, explains the choices, and sends users straight to where they can buy. With a database, current AI libraries and more planners, it can grow from this prototype into a full budgeting assistant."

11. Project Link

GitHub: <https://github.com/nandhinim25ds026-alt/PocketSmart-AI>

1. Project Scope

In scope

- User registration, login and logout with JWT authentication.
- Home, Party and Jewelry budget planners powered by Gemini.
- Optional outfit image upload for the jewelry planner.
- Shopping links for every recommended item.
- Dashboard with recent recommendations and a full history page.
- Documentation, test plan and demo for all eight phases.

Out of scope (future work)

- Permanent database, payments, live price tracking, mobile app, email features, shared plans.